

Theorem (Markov's Inequality). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow [0, \infty]$ be a non-negative random variable. For any real number $a > 0$,*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. **1. Indicator of the event $\{X \geq a\}$.** Define the indicator (characteristic) function

$$I_{\{X \geq a\}}(\omega) = \begin{cases} 1, & \text{if } X(\omega) \geq a, \\ 0, & \text{if } X(\omega) < a. \end{cases}$$

By definition, $I_{\{X \geq a\}}$ is a non-negative random variable taking only the values 0 and 1.

2. Pointwise inequality. Because $a > 0$ and $X(\omega) \geq 0$ for every ω ,

$$I_{\{X \geq a\}}(\omega) \leq \frac{X(\omega)}{a} \quad \text{for all } \omega \in \Omega. \quad (1)$$

Justification.

- If $X(\omega) \geq a$, then $I_{\{X \geq a\}}(\omega) = 1$ and $\frac{X(\omega)}{a} \geq \frac{a}{a} = 1$, so $1 \leq \frac{X(\omega)}{a}$.
- If $X(\omega) < a$, then $I_{\{X \geq a\}}(\omega) = 0$ while $\frac{X(\omega)}{a} \geq 0$; thus $0 \leq \frac{X(\omega)}{a}$.

Hence (1) holds in every case.

3. Monotonicity of expectation. The expectation operator is monotone for non-negative random variables: if $Y(\omega) \leq Z(\omega)$ for all ω , then $\mathbb{E}[Y] \leq \mathbb{E}[Z]$. Applying this to (1) gives

$$\mathbb{E}[I_{\{X \geq a\}}] \leq \mathbb{E}\left[\frac{X}{a}\right]. \quad (2)$$

4. Linearity of expectation. Since $\frac{X}{a} = \frac{1}{a} X$ and $\frac{1}{a}$ is a deterministic constant,

$$\mathbb{E}\left[\frac{X}{a}\right] = \frac{1}{a} \mathbb{E}[X]. \quad (3)$$

5. Expectation of an indicator equals a probability. For any event $A \in \mathcal{F}$, $\mathbb{E}[I_A] = \mathbb{P}(A)$; therefore

$$\mathbb{E}[I_{\{X \geq a\}}] = \mathbb{P}(X \geq a). \quad (4)$$

6. Combine the previous relations. Substituting (3) and (4) into (2) yields

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

which is precisely Markov's inequality.

7. Discussion of assumptions. The argument uses only that $X(\omega) \geq 0$ for all ω (so that X/a is non-negative) and that $a > 0$ (to avoid division by zero). No further integrability conditions are required beyond the existence (possibly infinite) of $\mathbb{E}[X]$.

□